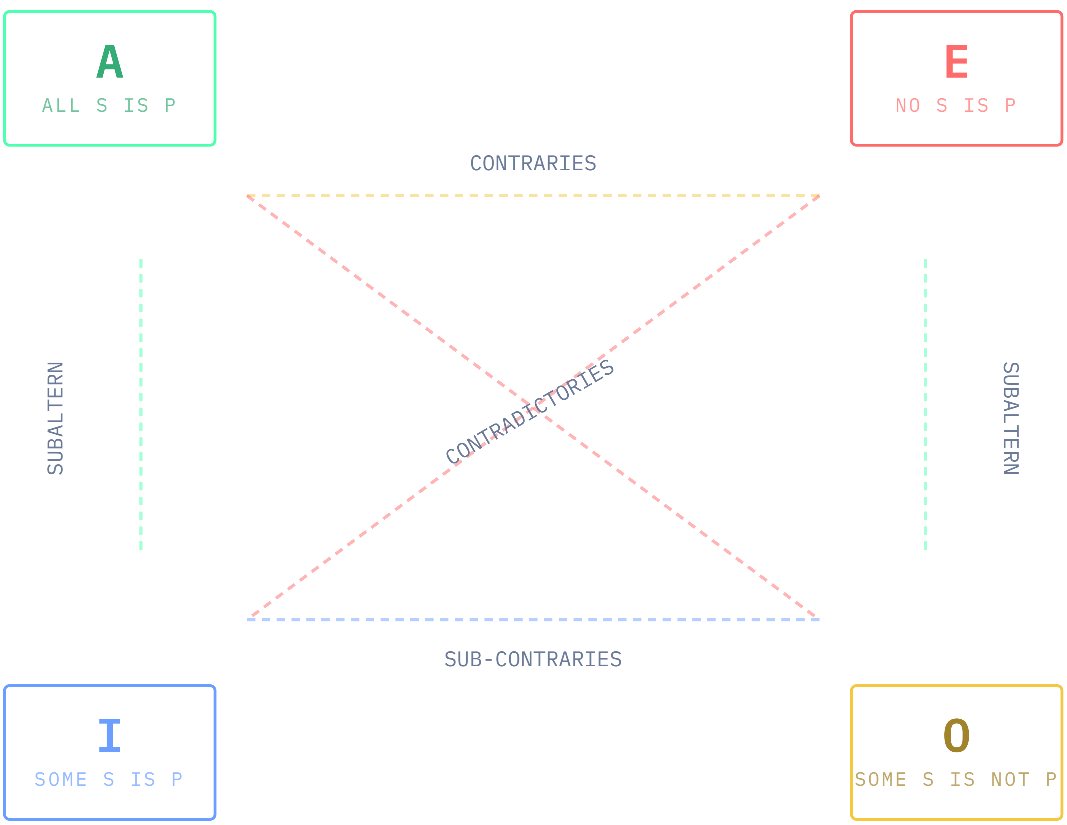
LOGIC · FORMAL ONTOLOGY · AI SYSTEMS

Knowledge Graph *Integrity & Conflict Detection*

An interactive system using the Square of Opposition and Porphyrian trees to detect contradictions, contraries, and taxonomic inconsistencies in knowledge graphs.

SQUARE OF OPPOSITION | ASSERTION CHECKER | PORPHYRIAN TREE

SQUARE OF OPPOSITION



Click any proposition node to explore its logical relationships and AI knowledge graph applications.

CONTRADICTORIES (A↔O, E↔I)

Cannot both be true AND cannot both be false. The strongest opposition – exactly one must hold. **Critical for AI:** flagging "All A is B" vs "Some A is not B" in a knowledge graph.

CONTRARIES (A↔E)

Cannot both be true, but can both be false. "All mammals fly" and "No mammals fly" – both false (bats fly, most don't). **Use:** detect impossible co-assertions in ontologies.

SUB-CONTRARIES (I↔O)

Cannot both be false, but can both be true. At least one partial claim must hold. Used to ensure minimum coverage in a taxonomy – a class must have some members in or out of a subclass.

SUBALTERNS (A→I, E→O)

Universal truths imply their particular counterparts. If "All S is P" is true, then "Some S is P" must be true. **Use:** inference propagation downward through a class hierarchy.